

JAVA PROGRAMMING FUNDAMENTALS

Experiment No: 01

Date: 16/09/26

Title: Student Utility Program using Java Fundamentals

Basic Information

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Objective / Aim

To design and implement a menu-driven Student Utility Program in Java using variables, data types, operators, conditional statements, loops, switch-case, methods, input/output, and basic problem-solving logic.

System Requirements

Hardware Requirements

- Processor: Intel Core i3 or equivalent
- RAM: 4 GB or higher
- Storage: At least 1 GB free space
- Keyboard and monitor

Software Requirements

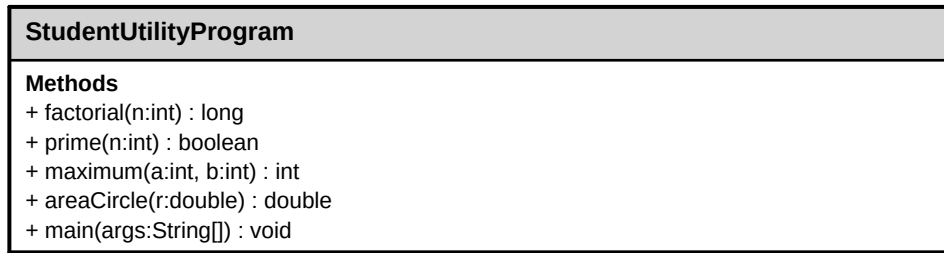
- Operating System: Windows / Linux / macOS
- JDK: Java Development Kit 17 or later (JDK 21 recommended)
- IDE: Eclipse / IntelliJ IDEA / NetBeans / VS Code
- Command Prompt / Terminal

Algorithm / Logic Description

1. Start the program and create a Scanner object for keyboard input.
2. Display the Student Utility Program menu containing 14 choices.
3. Read the user's menu choice using Scanner.
4. Use a switch statement to select the requested operation.
5. For Student Information, read name, roll number and three marks; calculate total and percentage.
6. For Even or Odd, use the modulus operator (%) to check divisibility by 2.
7. For Largest of Three, compare the three values using if-else conditions.
8. For Grade, compare the percentage against the specified grade ranges.
9. For Day of Week, use a nested switch to map numbers 1-7 to weekdays.
10. For the multiplication table and 1-to-N display, use a for loop.
11. For the sum of first N natural numbers, initialize sum to 0 and add values from 1 through N.
12. For Fibonacci, start with 0 and 1 and repeatedly generate the next term by adding the previous two terms.
13. For Factorial, call factorial(n), which multiplies values from 1 through n.
14. For Prime Number, call prime(n) and test divisibility from 2 through n/2.

15. For Maximum of Two Numbers, call `maximum(a,b)`, which uses the conditional operator.
16. For Area of Circle, call `areaCircle(r)` and calculate $3.14 \times r \times r$.
17. Repeat the menu using the do-while loop until the user selects option 14.
18. Display the exit message, close the Scanner object, and stop the program.

UML Class Diagram



The program contains one main class, StudentUtilityProgram. It contains four user-defined utility methods and the main() method. The main method creates a Scanner object and controls the menu, switch-case operations, loops, conditions, and method calls.

Source Code

```
import java.util.Scanner;

// Main class of the program
public class StudentUtilityProgram {

    // ----- Methods -----

    // Method to calculate factorial of a number
    public static long factorial(int n) {
        long fact = 1;

        // Multiply numbers from 1 to n
        for (int i = 1; i <= n; i++) {
            fact = fact * i;
        }

        return fact;
    }

    // Method to check whether a number is prime
    public static boolean prime(int n) {

        // Numbers less than or equal to 1 are not prime
        if (n <= 1)
            return false;

        // Check divisibility from 2 to n/2
        for (int i = 2; i <= n / 2; i++) {
            if (n % i == 0)
                return false;
        }

        return true;
    }

    // Method to find maximum of two numbers
    public static int maximum(int a, int b) {
        return (a > b) ? a : b;
    }

    // Method to calculate area of a circle
    public static double areaCircle(double r) {
        return 3.14 * r * r;
    }

    // Main method
    public static void main(String[] args) {

        // Create Scanner object to take input
        Scanner sc = new Scanner(System.in);

        int choice;

        // do-while loop displays the menu repeatedly
        do {

            // Display menu
            System.out.println("\n===== STUDENT UTILITY PROGRAM =====");
            System.out.println("1. Student Information");
            System.out.println("2. Even or Odd");
            System.out.println("3. Largest of Three Numbers");
            System.out.println("4. Grade using Percentage");
            System.out.println("5. Day of Week");
            System.out.println("6. Multiplication Table");
            System.out.println("7. Display 1 to N");
            System.out.println("8. Sum of First N Natural Numbers");
            System.out.println("9. Fibonacci Series");
            System.out.println("10. Factorial");
            System.out.println("11. Prime Number");
            System.out.println("12. Maximum of Two Numbers");
            System.out.println("13. Area of Circle");
            System.out.println("14. Exit");

            // Read user's choice
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();

            // Perform operation based on user's choice
            switch (choice) {

                // ----- Student Information -----
                case 1:

                    // Consume the leftover newline
                    sc.nextLine();

                    // Read student name
                    System.out.print("Enter Name: ");
                    String name = sc.nextLine();

                    // Read roll number
                    System.out.print("Enter Roll Number: ");
                    int roll = sc.nextInt();

                    // Read marks of three subjects
                    System.out.print("Enter Marks of Subject1: ");
                    int m1 = sc.nextInt();
```

```

        System.out.print("Enter Marks of Subject2: ");
        int m2 = sc.nextInt();

        System.out.print("Enter Marks of Subject3: ");
        int m3 = sc.nextInt();

        // Calculate total marks
        int total = m1 + m2 + m3;

        // Calculate percentage
        double percentage = total / 3.0;

        // Display student details
        System.out.println("Name : " + name);
        System.out.println("Roll No : " + roll);
        System.out.println("Total : " + total);
        System.out.println("Percentage : " + percentage);

        break;

// ----- Even or Odd -----
case 2:

    // Read a number
    System.out.print("Enter Number: ");
    int n = sc.nextInt();

    // Check whether number is divisible by 2
    if (n % 2 == 0)
        System.out.println("Even Number");
    else
        System.out.println("Odd Number");

    break;

// ----- Largest of Three Numbers -----
case 3:

    // Read three numbers
    System.out.print("Enter Three Numbers: ");
    int a = sc.nextInt();
    int b = sc.nextInt();
    int c = sc.nextInt();

    // Compare the three numbers
    if (a >= b && a >= c)
        System.out.println("Largest = " + a);
    else if (b >= c)
        System.out.println("Largest = " + b);
    else
        System.out.println("Largest = " + c);

    break;

// ----- Grade using Percentage -----
case 4:

    // Read percentage
    System.out.print("Enter Percentage: ");
    double per = sc.nextDouble();

    // Assign grade based on percentage
    if (per >= 90)
        System.out.println("Grade A");
    else if (per >= 75)
        System.out.println("Grade B");
    else if (per >= 60)
        System.out.println("Grade C");
    else if (per >= 40)
        System.out.println("Grade D");
    else
        System.out.println("Fail");

    break;

// ----- Day of Week -----
case 5:

    // Read day number
    System.out.print("Enter Day Number (1-7): ");
    int day = sc.nextInt();

    // Display day using switch
    switch (day) {

        case 1:
            System.out.println("Monday");
            break;

        case 2:
            System.out.println("Tuesday");
            break;

        case 3:
            System.out.println("Wednesday");
            break;

        case 4:
            System.out.println("Thursday");
            break;

        case 5:

```

```

        System.out.println("Friday");
        break;

    case 6:
        System.out.println("Saturday");
        break;

    case 7:
        System.out.println("Sunday");
        break;

    default:
        System.out.println("Invalid Day");
    }

    break;

// ----- Multiplication Table -----
case 6:

    // Read number
    System.out.print("Enter Number: ");
    int num = sc.nextInt();

    // Print multiplication table from 1 to 10
    for (int i = 1; i <= 10; i++) {
        System.out.println(
            num + " x " + i + " = " + (num * i)
        );
    }

    break;

// ----- Display 1 to N -----
case 7:

    // Read limit
    System.out.print("Enter N: ");
    int limit = sc.nextInt();

    // Display numbers from 1 to N
    for (int i = 1; i <= limit; i++) {
        System.out.print(i + " ");
    }

    System.out.println();
    break;

// ----- Sum of First N Natural Numbers -----
case 8:

    // Read N
    System.out.print("Enter N: ");
    int x = sc.nextInt();

    int sum = 0;

    // Calculate sum from 1 to N
    for (int i = 1; i <= x; i++)
        sum += i;

    // Display sum
    System.out.println("Sum = " + sum);

    break;

// ----- Fibonacci Series -----
case 9:

    // Read number of terms
    System.out.print("Enter Number of Terms: ");
    int terms = sc.nextInt();

    // First two Fibonacci numbers
    int first = 0, second = 1;

    // Display first two numbers
    System.out.print(first + " " + second + " ");

    // Generate remaining Fibonacci numbers
    for (int i = 3; i <= terms; i++) {
        int next = first + second;

        System.out.print(next + " ");

        // Update values for next iteration
        first = second;
        second = next;
    }

    System.out.println();
    break;

// ----- Factorial -----
case 10:

    // Read number
    System.out.print("Enter Number: ");
    int fact = sc.nextInt();
    // Call factorial method

```

```

        System.out.println("Factorial = " + factorial(fact));

        break;

// ----- Prime Number -----
case 11:

    // Read number
    System.out.print("Enter Number: ");
    int p = sc.nextInt();

    // Call prime method and check result
    if (prime(p))
        System.out.println("Prime Number");
    else
        System.out.println("Not Prime");

    break;

// ----- Maximum of Two Numbers -----
case 12:

    // Read two numbers
    System.out.print("Enter Two Numbers: ");
    int n1 = sc.nextInt();
    int n2 = sc.nextInt();

    // Call maximum method
    System.out.println("Maximum = " + maximum(n1, n2));

    break;

// ----- Area of Circle -----
case 13:

    // Read radius
    System.out.print("Enter Radius: ");
    double r = sc.nextDouble();

    // Call areaCircle method
    System.out.println("Area = " + areaCircle(r));

    break;

// ----- Exit -----
case 14:

    // Exit message
    System.out.println("Thank You");
    break;

// Handle invalid menu choice
default:
    System.out.println("Invalid Choice");
}

// Continue until user chooses option 14
} while (choice != 14);

// Close Scanner object
sc.close();
}
}

```

Compilation and Execution Commands

Save the program using the exact file name **StudentUtilityProgram.java**.

Compilation:

```
javac StudentUtilityProgram.java
```

Execution:

```
java StudentUtilityProgram
```

If there are no compilation errors, the execution command starts the menu-driven program.

Output

Sample Output 1 – Student Information

```
===== STUDENT UTILITY PROGRAM =====
1. Student Information
2. Even or Odd
3. Largest of Three Numbers
4. Grade using Percentage
5. Day of Week
6. Multiplication Table
7. Display 1 to N
8. Sum of First N Natural Numbers
9. Fibonacci Series
10. Factorial
11. Prime Number
12. Maximum of Two Numbers
13. Area of Circle
14. Exit
Enter your choice: 1
Enter Name: Rahul
Enter Roll Number: 101
Enter Marks of Subject1: 80
Enter Marks of Subject2: 90
Enter Marks of Subject3: 70
Name : Rahul
Roll No : 101
Total : 240
Percentage : 80.0
```

Sample Output 2 – Factorial

```
Enter your choice: 10
Enter Number: 5
Factorial = 120
```

Sample Output 3 – Prime Number

```
Enter your choice: 11
Enter Number: 17
Prime Number
```

Sample Output 4 – Exit

```
Enter your choice: 14
Thank You
```

Result

The Student Utility Program was designed and implemented successfully using Java programming fundamentals, including input/output, conditional statements, loops, switch-case, methods, and a menu-driven do-while structure.